

NUMERICAL ANALYSIS

PROBLEM SET 5

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1. PROBLEM 1

Derive the trapezoid integration scheme as an Adams-Moulton method with $m = 1$ step:

$$\frac{w_{j+1} - w_j}{h} = \frac{1}{2}[f(t_{j+1}, w_{j+1}) + f(t_j, w_j)]$$

to approximate the initial value problem $y' = f(t, y)$ for $a \leq t \leq b$ with initial condition $y(a) = \alpha$. Also derive the associated error term. Let $w_j \approx y_j := y(t_j)$.

- (a) Use the Lagrange form of the polynomial interpolant to write

$$f(t, y(t)) = P_1(t) + R_1(t).$$

Integrate $P_1(t) = L_{1,0}(t)f(t_{j+1}, y_{j+1}) + L_{1,1}(t)f(t_j, y_j)$ from $t = t_j$ to $t = t_j + 1$.

- (b) Set $w_{j+1} = w_j + \int_{t_j}^{t_{j+1}} f(t, y(t))dt$, approximating the 2nd term with $\int_{t_j}^{t_{j+1}} P_1(t)dt$, and derive the method above.
- (c) The error term $R_1(t) = \frac{f''(\xi, y(\xi))}{2}(t - t_{j+1})(t - t_j)$ for f (some $\xi \in (t_j, t_{j+1})$) can be integrated from $t = t_j$ to $t = t_{j+1}$ to get the error associated with the Adams-Moulton scheme:

$$\text{Error} = Mh^3 f''(\mu, y(\mu)).$$

Do this integration to find value of M .

- (d) What is the order of this one-step implicit method?

Solution. For the sake of convenience, we're going to set $g(t) = f(y, y(t)) = y'(t)$, and furthermore, we assume a uniform grid over which we will work. So set

$$t_j = a + jh, \quad h = \frac{b - a}{N}.$$

So we have that

$$y(t_{j+1} - y(t_j)) = \int_{t_j}^{t_{j+1}} g(t)dt.$$

- (a) So if we are to interpolate the function $g(t) = f(t, y(t))$ at the nodes t_{j+1} and t_j , then the degree-one Lagrange interpolant follows as $P_1(t) = L_{1,0}(t)f(t_{j+1}, y_{j+1}) + L_{1,1}(t)f(t_j, y_j)$,

where

$$L_{1,0}(t) = \frac{t - t_j}{t_{j+1} - t_j} = \frac{t - t_j}{h},$$

and

$$L_{1,1}(t) = \frac{t - t_{j+1}}{t_j - t_{j+1}} = \frac{t_{j+1} - t}{h}.$$

Thus, we have that

$$f(t, y(t)) = P_1(t) + R_1(t)$$

in conjunction with

$$P_1(t) = \frac{t - t_j}{h} f(t_{j+1}, y_{j+1}) + \frac{t_{j+1} - t}{h} f(t_j, y_j).$$

Integrating over our interval $[t_j, t_{j+1}]$ yields

$$\int_{t_j}^{t_{j+1}} P_1(t) dt = f(t_{j+1}, y_{j+1}) \int_{t_j}^{t_{j+1}} \frac{t - t_j}{h} dt + f(t_j, y_j) \int_{t_j}^{t_{j+1}} \frac{t_{j+1} - t}{h} dt.$$

(b) Starting from the integral form of the differential equation,

$$y_{j+1} = y_j + \int_{t_j}^{t_{j+1}} f(t, y(t)) dt.$$

Using the degree-one interpolating polynomial $P_1(t)$ from part (a), we approximate

$$\int_{t_j}^{t_{j+1}} f(t, y(t)) dt \approx \int_{t_j}^{t_{j+1}} P_1(t) dt.$$

From part (a),

$$\int_{t_j}^{t_{j+1}} P_1(t) dt = \frac{h}{2} [f(t_{j+1}, y_{j+1}) + f(t_j, y_j)].$$

Therefore,

$$y_{j+1} \approx y_j + \frac{h}{2} [f(t_{j+1}, y_{j+1}) + f(t_j, y_j)].$$

Replacing the exact values y_j and y_{j+1} by the numerical approximations w_j and w_{j+1} , we obtain

$$w_{j+1} = w_j + \frac{h}{2} [f(t_{j+1}, w_{j+1}) + f(t_j, w_j)].$$

Equivalently,

$$\frac{w_{j+1} - w_j}{h} = \frac{1}{2} [f(t_{j+1}, w_{j+1}) + f(t_j, w_j)].$$

This is the one-step Adams–Moulton method, also called the trapezoidal method.

(c) The interpolation remainder is

$$R_1(t) = \frac{f''(\xi, y(\xi))}{2} (t - t_{j+1})(t - t_j),$$

for some $\xi \in (t_j, t_{j+1})$. Hence the error in the integral approximation is

$$E = \int_{t_j}^{t_{j+1}} R_1(t) dt.$$

By the mean value theorem for integrals, for some $\mu \in (t_j, t_{j+1})$,

$$E = \frac{f''(\mu, y(\mu))}{2} \int_{t_j}^{t_{j+1}} (t - t_{j+1})(t - t_j) dt.$$

Let

$$s = t - t_j.$$

Since $t_{j+1} = t_j + h$, we have

$$t - t_j = s, \quad t - t_{j+1} = s - h,$$

and the limits become $0 \leq s \leq h$. Thus

$$\begin{aligned} \int_{t_j}^{t_{j+1}} (t - t_{j+1})(t - t_j) dt &= \int_0^h (s - h)s ds \\ &= \int_0^h (s^2 - hs) ds \\ &= \left[\frac{s^3}{3} - \frac{hs^2}{2} \right]_0^h \\ &= \frac{h^3}{3} - \frac{h^3}{2} \\ &= -\frac{h^3}{6}. \end{aligned}$$

Therefore,

$$E = \frac{f''(\mu, y(\mu))}{2} \left(-\frac{h^3}{6} \right) = -\frac{h^3}{12} f''(\mu, y(\mu)).$$

Thus, in the form

$$\text{Error} = Mh^3 f''(\mu, y(\mu)),$$

we have

$$M = -\frac{1}{12}.$$

(d) The local truncation error is proportional to h^3 :

$$E = O(h^3).$$

Therefore, the accumulated global error is $O(h^2)$. Hence the one-step implicit trapezoidal Adams–Moulton method is a second-order method. $\square$

2. PROBLEM 2

Consider the initial value problem

$$y' = \frac{4t\sqrt{1+y^2}}{y}, y(0) = 1$$

for $0 \leq t \leq 5$.

Challenge. Show that $y(t) = \sqrt{(2t^2 + \sqrt{2})^2 - 1}$ is a solution to the initial value problem. Is it unique?

Numerically implement the following methods below, and run with enough values of step-size h to demonstrate that each of has convergence of global discretization error with rate $\mathcal{O}(h^2)$.

- (a) Optimal RK2 method
- (b) Two-step Adams-Bashforth
- (c) One-step Adams-Moulton method
- (d) Heun method (predictor-corrector)

Plot the error versus the step size on a log-log plot for all methods on the same axes to compare. Comment on the advantages and disadvantages of each.

Solution. Define

$$f(t, y) = \frac{4t\sqrt{1+y^2}}{y}, \quad y_{\text{exact}}(t) = \sqrt{(2t^2 + \sqrt{2})^2 - 1}.$$

For a step size $h = 5/N$, let $t_n = nh$ and let w_n approximate $y(t_n)$. The global discretization error is measured by

$$E(h) = \max_{0 \leq n \leq N} |w_n - y_{\text{exact}}(t_n)|.$$

The step sizes used were

$$h = 0.2, 0.1, 0.05, 0.025, 0.0125, 0.00625.$$

(a) *Optimal RK2 method.* We use Ralston's optimal two-stage RK2 method:

$$\begin{aligned} k_1 &= f(t_n, w_n), \\ k_2 &= f\left(t_n + \frac{2h}{3}, w_n + \frac{2h}{3}k_1\right), \\ w_{n+1} &= w_n + h\left(\frac{1}{4}k_1 + \frac{3}{4}k_2\right). \end{aligned}$$

This is an explicit, self-starting, second-order method.

(b) *Two-step Adams–Bashforth method.* For $n \geq 1$, the method is

$$w_{n+1} = w_n + \frac{h}{2} (3f(t_n, w_n) - f(t_{n-1}, w_{n-1})).$$

Since this method requires two previous values, w_1 was computed using one Ralston RK2 step. A second-order starting method is necessary to preserve the second-order global accuracy of Adams–Bashforth.

(c) *One-step Adams–Moulton method.* The one-step Adams–Moulton method is the implicit trapezoidal rule

$$w_{n+1} = w_n + \frac{h}{2} [f(t_n, w_n) + f(t_{n+1}, w_{n+1})].$$

At each step, w_{n+1} was found by solving

$$G(z) = z - w_n - \frac{h}{2} [f(t_n, w_n) + f(t_{n+1}, z)] = 0$$

with Newton’s method. The initial Newton iterate was the forward-Euler prediction

$$z^{(0)} = w_n + hf(t_n, w_n).$$

Since

$$f_y(t, y) = -\frac{4t}{y^2\sqrt{1+y^2}},$$

the Newton iteration is

$$z^{(j+1)} = z^{(j)} - \frac{G(z^{(j)})}{1 + \frac{2h(t_n + h)}{(z^{(j)})^2\sqrt{1+(z^{(j)})^2}}}.$$

(d) *Heun predictor–corrector method.* First, a forward-Euler predictor is computed:

$$\tilde{w}_{n+1} = w_n + hf(t_n, w_n).$$

One trapezoidal correction then gives

$$w_{n+1} = w_n + \frac{h}{2} [f(t_n, w_n) + f(t_{n+1}, \tilde{w}_{n+1})].$$

This is an explicit method because the corrector evaluates f at the predicted value rather than at the unknown corrected value.

The MATLAB implementation is contained in `ps5prob2.m`. It may be included in the document using

`\lstinputlisting{ps5prob2.m}.`

Equivalently, the following command can be placed outside display mode:

LISTING 1. MATLAB implementation for Problem 2.

<pre>%% Problem 2: second-order convergence of four numerical ODE methods</pre>

```

% Global error is max_n |y_n - y_exact(t_n)| on 0 <= t <= 5.

clear; clc; close all;

T = 5;
Nvalues = [25, 50, 100, 200, 400, 800];
hvalues = T ./ Nvalues;

methodNames = { ...
    'Optimal_RK2_(Ralston)', ...
    'Adams-Bashforth_2', ...
    'Adams-Moulton_1', ...
    'Heun_predictor-corrector'};

errors = zeros(numel(methodNames), numel(hvalues));
rates = NaN(size(errors));

for m = 1:numel(methodNames)
    for j = 1:numel(hvalues)
        h = hvalues(j);

        switch m
            case 1
                [t, y] = solveOneStep(@ralstonStep, T, h);
            case 2
                [t, y] = solveAB2(T, h);
            case 3
                [t, y] = solveOneStep(@adamsMoultonStep, T, h);
            case 4
                [t, y] = solveOneStep(@heunStep, T, h);
        end

        errors(m,j) = max(abs(y - yExact(t)));
        if j > 1
            rates(m,j) = log(errors(m,j-1)/errors(m,j))/log(2);
        end
    end
end

%% Print convergence tables
for m = 1:numel(methodNames)
    fprintf('\n%s\n', methodNames{m});
    fprintf('%10s_16s_10s\n', 'h', 'max_error', 'rate');
    fprintf('%10.6f_16.8e_10s\n', hvalues(1), errors(m,1), '--');
    for j = 2:numel(hvalues)
        fprintf('%10.6f_16.8e_10.4f\n', ...
            hvalues(j), errors(m,j), rates(m,j));
    end
end

```

```

        end
    end

    %% Plot every method on the same log-log axes
    figure('Color', 'w');
    hold on;
    styles = {'o-', 's-', 'd-', '~-'};

    for m = 1:numel(methodNames)
        loglog(hvalues, errors(m,:), styles{m}, ...
            'LineWidth', 1.5, 'MarkerSize', 7, ...
            'DisplayName', methodNames{m});
    end

    % A line proportional to h^2 for comparison.
    reference = errors(1,end) .* (hvalues./hvalues(end)).^2;
    loglog(hvalues, reference, 'k--', 'LineWidth', 1.3, ...
        'DisplayName', 'reference_h^2');

    grid on;
    xlabel('Step_size_h');
    ylabel('Maximum_nodal_error');
    title('Second-order_convergence_on_0_leq_t_leq_5');
    legend('Location', 'northwest');
    set(gca, 'FontSize', 11);
    exportgraphics(gcf, 'problem2_convergence_matlab.png', 'Resolution', 200);

    %% Local functions
    function value = rhs(t, y)
        value = 4*t*sqrt(1 + y.^2)./y;
    end

    function value = yExact(t)
        value = sqrt((2*t.^2 + sqrt(2)).^2 - 1);
    end

    function yNext = ralstonStep(t, y, h)
        % Optimal two-stage, second-order Runge-Kutta method.
        k1 = rhs(t, y);
        k2 = rhs(t + 2*h/3, y + 2*h*k1/3);
        yNext = y + h*(k1/4 + 3*k2/4);
    end

    function yNext = heunStep(t, y, h)
        % Euler predictor followed by one trapezoidal correction.
        predictor = y + h*rhs(t, y);
        yNext = y + (h/2)*(rhs(t, y) + rhs(t + h, predictor));
    end

```

```

end

function yNext = adamsMoultonStep(t, y, h)
    % One-step Adams-Moulton (implicit trapezoidal rule).
    % Newton iteration solves
    %  $z - y - h/2*(f(t,y) + f(t+h,z)) = 0$ .
    fn = rhs(t, y);
    z = y + h*fn; % Euler initial guess
    tolerance = 1e-13;
    maximumIterations = 20;

    for iteration = 1:maximumIterations
        residual = z - y - (h/2)*(fn + rhs(t + h, z));

        % Since  $d/dz[\sqrt{1+z^2}/z] = -1/(z^2\sqrt{1+z^2})$ ,
        % this is the derivative of the residual with respect to z.
        jacobian = 1 + 2*h*(t + h)/(z^2*sqrt(1 + z^2));
        correction = residual/jacobian;
        z = z - correction;

        if abs(correction) <= tolerance*max(1, abs(z))
            yNext = z;
            return;
        end
    end

    error('Newton iteration did not converge.');
```

```

end

function [t, y] = solveOneStep(stepFunction, T, h)
    N = round(T/h);
    t = (0:N)*h;
    y = zeros(size(t));
    y(1) = 1;

    for n = 1:N
        y(n+1) = stepFunction(t(n), y(n), h);
    end
end

function [t, y] = solveAB2(T, h)
    N = round(T/h);
    t = (0:N)*h;
    y = zeros(size(t));
    y(1) = 1;

    % A second-order startup is required to retain AB2's global order.

```



```

y(2) = ralstonStep(t(1), y(1), h);

for n = 2:N
    fn = rhs(t(n), y(n));
    fnMinus1 = rhs(t(n-1), y(n-1));
    y(n+1) = y(n) + h*(3*fn - fnMinus1)/2;
end
end

%=====OUTPUT=====
%Optimal RK2 (Ralston)
%           h           max error           rate
% 0.200000    8.92440542e-03           --
% 0.100000    2.08787877e-03          2.0957
% 0.050000    5.09660918e-04          2.0344
% 0.025000    1.26166968e-04          2.0142
% 0.012500    3.14027146e-05          2.0064
% 0.006250    7.83433508e-06          2.0030
%
%Adams-Bashforth 2
%           h           max error           rate
% 0.200000    2.88534227e-02           --
% 0.100000    7.22165611e-03          1.9983
% 0.050000    1.80363844e-03          2.0014
% 0.025000    4.50467567e-04          2.0014
% 0.012500    1.12546757e-04          2.0009
% 0.006250    2.81276299e-05          2.0005
%
%Adams-Moulton 1
%           h           max error           rate
% 0.200000    5.87539115e-03           --
% 0.100000    1.44663578e-03          2.0220
% 0.050000    3.60330581e-04          2.0053
% 0.025000    9.00004500e-05          2.0013
% 0.012500    2.24949879e-05          2.0003
% 0.006250    5.62376734e-06          2.0000
%
%Heun predictor-corrector
%           h           max error           rate
% 0.200000    9.20480095e-03           --
% 0.100000    2.10736388e-03          2.1269
% 0.050000    5.11123795e-04          2.0437
% 0.025000    1.26238053e-04          2.0175
% 0.012500    3.13904354e-05          2.0077
% 0.006250    7.82788897e-06          2.0036
%>>

```

The observed maximum nodal errors were as follows:

h	Ralston RK2	AB2	AM1	Heun
0.20000	8.9244×10^{-3}	2.8853×10^{-2}	5.8754×10^{-3}	9.2048×10^{-3}
0.10000	2.0879×10^{-3}	7.2217×10^{-3}	1.4466×10^{-3}	2.1074×10^{-3}
0.05000	5.0966×10^{-4}	1.8036×10^{-3}	3.6033×10^{-4}	5.1112×10^{-4}
0.02500	1.2617×10^{-4}	4.5047×10^{-4}	9.0000×10^{-5}	1.2624×10^{-4}
0.01250	3.1403×10^{-5}	1.1255×10^{-4}	2.2495×10^{-5}	3.1390×10^{-5}
0.00625	7.8343×10^{-6}	2.8128×10^{-5}	5.6238×10^{-6}	7.8279×10^{-6}

The experimental convergence rate is calculated from

$$p(h) = \log_2 \left(\frac{E(h)}{E(h/2)} \right).$$

At the smallest step sizes, the observed rates were

Method	$p(h)$
Optimal RK2	2.0030
Two-step Adams–Bashforth	2.0005
One-step Adams–Moulton	2.0000
Heun predictor–corrector	2.0036

Thus, halving h decreases the global error by approximately a factor of four for every method. This demonstrates

$$E(h) = \mathcal{O}(h^2)$$

for all four schemes.

Figure 1 compares the errors of the four numerical methods with a reference curve proportional to h^2 . The computed rates approaching 2 confirm second-order global convergence.

Advantages and disadvantages. Ralston’s RK2 method is explicit and self-starting, but it requires two function evaluations per step and has the stability limitations of an explicit method. The two-step Adams–Bashforth method requires only one new function evaluation per step after startup, but it is not self-starting, must store a previous value, and has the largest error constant in this experiment.

The one-step Adams–Moulton method gives the smallest errors and has better stability properties, but it requires a nonlinear solve at every step. Heun’s method is explicit, self-starting, and easy to implement, but it requires two function evaluations per step and generally has weaker stability properties than the implicit Adams–Moulton method. $\square$

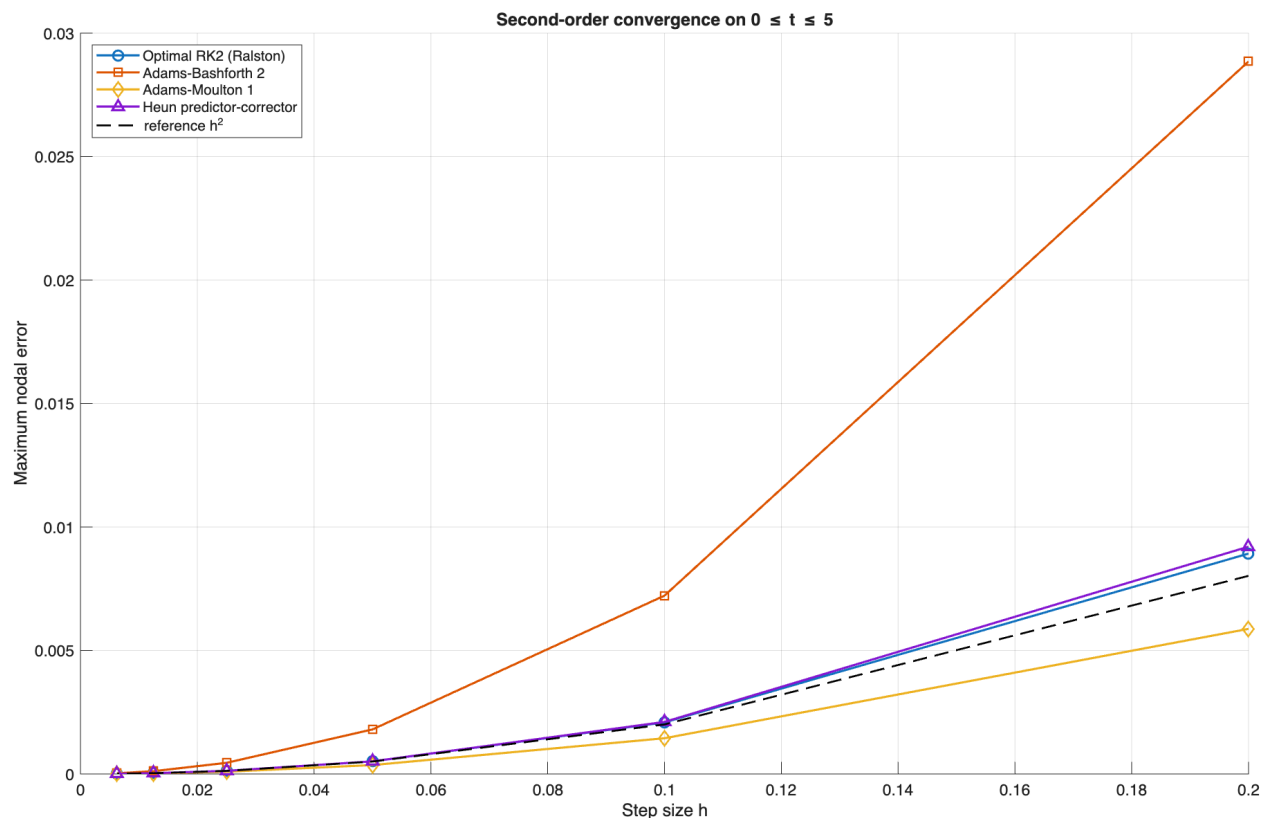


FIGURE 1. Maximum nodal error versus step size for the optimal RK2, two-step Adams–Bashforth, one-step Adams–Moulton, and Heun predictor–corrector methods. The dashed curve is proportional to h^2 .

3. PROBLEM 3

Consider the linear multi-step method to approximate the initial value problem $y' = f(t, y)$ for $a \leq t \leq b$ with the initial condition $y(a) = \alpha$:

$$w_{j+1} - \frac{18}{11}w_j + \frac{9}{11}w_{j-1} + \frac{2}{11}w_{j-2} = \frac{6}{11}hf(t_{j+1}, w_{j+1}).$$

- Is the method stable? If so, is it strongly or weakly stable? Show your work by considering the roots of the characteristic polynomial, noting that $\lambda = 1$ is a root, and you can even factor and write $P(\lambda) = (\lambda - 1) * (\dots)$.
- Is the method convergent?
- Is the method consistent?

Solution. Consider the linear multistep method

$$w_{j+1} - \frac{18}{11}w_j + \frac{9}{11}w_{j-1} - \frac{2}{11}w_{j-2} = \frac{6}{11}hf(t_{j+1}, w_{j+1}).$$

- (a) To determine stability, we consider the characteristic polynomial associated with the homogeneous difference equation:

$$P(\lambda) = \lambda^3 - \frac{18}{11}\lambda^2 + \frac{9}{11}\lambda - \frac{2}{11}.$$

Multiplying by 11,

$$11P(\lambda) = 11\lambda^3 - 18\lambda^2 + 9\lambda - 2.$$

Since $\lambda = 1$ is a root, we factor:

$$11\lambda^3 - 18\lambda^2 + 9\lambda - 2 = (\lambda - 1)(11\lambda^2 - 7\lambda + 2).$$

Thus,

$$P(\lambda) = (\lambda - 1) \left(\lambda^2 - \frac{7}{11}\lambda + \frac{2}{11} \right).$$

The remaining roots satisfy

$$11\lambda^2 - 7\lambda + 2 = 0.$$

Using the quadratic formula,

$$\lambda = \frac{7 \pm \sqrt{49 - 88}}{22} = \frac{7 \pm i\sqrt{39}}{22}.$$

Their magnitudes are

$$|\lambda| = \sqrt{\left(\frac{7}{22}\right)^2 + \left(\frac{\sqrt{39}}{22}\right)^2} = \frac{\sqrt{88}}{22} = \frac{\sqrt{22}}{11} < 1.$$

Therefore, the three roots are

$$\lambda_1 = 1, \quad \lambda_{2,3} = \frac{7 \pm i\sqrt{39}}{22}.$$

All roots satisfy

$$|\lambda_k| \leq 1,$$

and the only root on the unit circle is the simple root $\lambda = 1$. Hence the root condition is satisfied, so the method is stable.

Furthermore, since all roots other than the principal root $\lambda = 1$ lie strictly inside the unit circle, the method is

strongly stable.

- (b) A linear multistep method is convergent if and only if it is both consistent and stable (zero-stable).

From part (a), the method satisfies the root condition and is therefore stable. We verify consistency in part (c). Since the method is both stable and consistent, it

follows that

the method is convergent.

(c) For a linear multistep method, define

$$\rho(\lambda) = \lambda^3 - \frac{18}{11}\lambda^2 + \frac{9}{11}\lambda - \frac{2}{11},$$

and

$$\sigma(\lambda) = \frac{6}{11}\lambda^3.$$

A linear multistep method is consistent if

$$\rho(1) = 0$$

and

$$\rho'(1) = \sigma(1).$$

First,

$$\rho(1) = 1 - \frac{18}{11} + \frac{9}{11} - \frac{2}{11} = \frac{11 - 18 + 9 - 2}{11} = 0.$$

Next,

$$\rho'(\lambda) = 3\lambda^2 - \frac{36}{11}\lambda + \frac{9}{11}.$$

Therefore,

$$\rho'(1) = 3 - \frac{36}{11} + \frac{9}{11} = \frac{33 - 36 + 9}{11} = \frac{6}{11}.$$

Also,

$$\sigma(1) = \frac{6}{11}.$$

Hence,

$$\rho'(1) = \sigma(1).$$

Both consistency conditions are satisfied, and therefore

the method is consistent.

□

4. PROBLEM 4

Use a built-in ODE solver to numerically integrate the Lorenz equations

$$\begin{aligned}\dot{x} &= \sigma(x - y) \\ \dot{y} &= rx - y - xz \\ \dot{z} &= xy - bz\end{aligned}$$

with the following values for coefficients $\sigma = 10, r = 28, b = 8/3$.

- (a) Using the initial condition $(x_0, y_0, z_0) = (5, 5, 5)$, solve the system for $0 \leq t \leq 10$ with a time step $h = 0.001$. Make a 2D plot of x versus z and a 3D plot of x versus y versus z .
- (b) Use a new initial condition $(\bar{x}_0, \bar{y}_0, \bar{z}_0) = (5, 5, 5)10^{-5}$, repeat the simulations in part (a) and the plots in part (b).
- (c) Plot the L_2 -norm of the difference in the solutions obtained using the different initial conditions in (a) and (b). Comment on your results.
- (d) Repeat parts (a) - (c) with different solvers and different step sizes. Comment on your results. You may see differences when using some solvers but not using others. Given the differences observed when using different methods, how can you tell what to expect for the exact solution?

Solution.

□